



SIMATS ENGINEERING

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Course Code: CSA0711

Slot: D

Course Name: Computer Networks

Course Faculty: Dr. Shankar

Project Title: Artificial Intelligence for Smart Traffic Prediction and Optimization in Next-Generation Networks

Module Photographs:



MODULE 3: TRAFFIC OPTIMIZATION AND RESOURCE MANAGEMENT

- Select the best routing path based on AI predictions.
- Dynamically allocate bandwidth and network resources.
- Reduce network congestion, latency, and packet loss.
- Improve throughput, Quality of Service (QoS), and overall network performance.

Project Description:

The Artificial Intelligence for Smart Traffic Prediction and Optimization in Next-Generation Networks project is designed to provide an intelligent system for monitoring, predicting, and managing network traffic. The project collects real-time and historical network traffic data and preprocesses it by removing missing or noisy data and extracting important traffic features. Machine Learning and Deep Learning techniques are used to analyze traffic patterns and predict future network conditions. The system identifies possible network congestion in advance and uses AI-based predictions to support dynamic traffic routing and efficient resource allocation. Based on the predicted traffic conditions, the system selects suitable network paths, manages bandwidth, and supports load balancing to reduce congestion, latency, and packet loss. Real-time network monitoring and congestion detection help the system respond effectively to continuously changing traffic conditions. The proposed system focuses on improving network throughput, bandwidth utilization, reliability, and Quality of Service (QoS). It is suitable for next-generation communication environments such as 5G/6G networks, IoT systems, cloud services, smart cities, video streaming, and online applications that require fast and reliable communication. This project demonstrates the practical application of Artificial Intelligence, Machine Learning, Deep Learning, network traffic analysis, dynamic routing, and resource management for improving the efficiency and performance of next-generation computer networks.

Student Signature

Guide Signature